CLARITHROMYCIN

Strong CYP3A4 inhibitor
QT prolongation

CHEMICAL NAME 6-O-methylerythromycin
CLASS Macrolide antibiotic
TRADE NAMES Klacid · Biaxin · Klaricid · Claridar

THERAPEUTIC USES / INDICATIONS

- Respiratory tract infections: CAP (atypical cover), sinusitis, pharyngitis, bronchitis.
- *H. pylori* eradication (triple therapy with PPI + amoxicillin/metronidazole).
- Skin and soft-tissue infections (penicillin allergy).
- *Mycobacterium avium* complex.

ADMINISTRATION

- **PO** tablets (MR with food), suspension.
- **IV** infusion over 60 min into a large vein — never bolus or IM.
- Check interacting drugs before starting.

DOSAGES

- **Adult:** 250–500 mg every 12 h for 7–14 days.
- **H. pylori:** 500 mg every 12 h for 7–14 days.
- **IV:** 500 mg every 12 h.
- **Children:** 7.5 mg/kg every 12 h (max 500 mg/dose).
- CrCl <30: halve dose.

PHARMACOKINETICS

- **Peak:** 2 h. Bioavailability ~50%.
- Metabolised in liver (CYP3A4) to active 14-hydroxy metabolite.
- Excreted in urine and faeces. **t½:** 3–7 h.

PHARMACODYNAMICS

- Binds **50S ribosomal subunit** → inhibits protein synthesis.
- Better acid stability and activity than erythromycin; active against *H. pylori*.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Taste disturbance (metallic), nausea, diarrhoea.
- Abdominal pain, dyspepsia.
- Headache, insomnia.

ADVERSE EFFECTS (SERIOUS)

- QT prolongation, ventricular arrhythmias.
- Hepatotoxicity.
- *C. difficile* colitis.
- ↑ cardiovascular mortality in patients with heart disease (long-term).

CONTRAINDICATIONS & CAUTIONS

- Macrolide hypersensitivity.
- Use with statins (simvastatin, lovastatin), colchicine (in renal/hepatic impairment), ergotamine, pimozide, ticagrelor; dangerous interactions.
- Long QT, hypokalaemia; severe hepatic failure with renal impairment.

KEY INTERACTIONS

- **Strong CYP3A4 inhibition:** ↑ levels of simvastatin (rhabdomyolysis), colchicine (fatal), carbamazepine, ciclosporin, tacrolimus, midazolam, calcium-channel blockers.
- **Warfarin:** ↑ INR.
- **QT-prolonging drugs:** additive.

NURSING MANAGEMENT

- Review all medications for interactions before giving.
- Hold simvastatin during short courses if instructed.
- Check ECG and electrolytes in at-risk patients.
- IV: large vein, 60 min infusion; monitor site.
- Monitor LFTs, GI symptoms.

PATIENT EDUCATION

- Metallic taste is common.
- Tell your doctor/pharmacist about all medicines (especially statins).
- Report palpitations, fainting, yellow skin.
- Complete the course.

KEY POINTS

- Macrolide used in H. pylori triple therapy.
- QT prolongation.
- Strong CYP3A4 inhibitor — many interactions.
- Never IV bolus.